IAA - Deliverable 1

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1 Introduction

This project aims to develop a data-driven AI player for the card game Sueca, using supervised machine learning to improve decision-making during play. The goal then becomes to improve decision-making during the player turn using machine learning.

2 Context and Motivation

Sueca is a card game that requires complex decision-making under hidden information, where players must balance individual strategy with team coordination. Traditional rule-based or heuristic AI approaches struggle to capture this complexity and often fail to adapt to diverse game situations. This motivates the use of machine learning to learn decision patterns from data, with the potential to improve the quality of plays and enable applications that require a competitive game AI, strategy analysis, and large-scale simulations.

2.1 Data Source

The dataset will be created specifically for this project by running large-scale simulations of Sueca games using several advanced AI bots. In these simulations, four automated players compete in self-play, and every decision, trick outcome, and final game result is logged.

Optionally, the dataset may later be extended with logs from human games, but the initial focus will be on synthetic data generated by AI agents.

2.2 Data Size and Scope

The goal is to generate data from a large number of full games, covering a wide variety of situations (different trumps, score states, and player positions).

By logging every decision in each simulated game, we expect to obtain from tens of thousands to hundreds of thousands of decision instances, providing sufficient coverage of typical play scenarios.

2.3 Data Structure

Each dataset instance will correspond to a decision point in the game, i.e., the moment when a player has to choose which card to play.

The input features will encode the game state from the acting player's perspective, including:

- **Hand representation**: all cards currently held by the player, identifying suit, rank, and whether each card is a trump.
- **Current trick**: lead suit, cards already played in the current trick, and which card is currently winning the trick.
- **Trump suit**: the suit defined as trump for the current game.
- **Previous plays**: summary of past tricks (e.g., cards already played, number of tricks won by each team, points accumulated so far).
- **Player position**: the player's seat at the table (first, second, third, or fourth to play in the trick) and team membership.
- **Game progress**: round index, number of remaining cards, and possibly the score difference between teams.

The target will be derived from the behaviour of the advanced bots:

- In the simplest formulation, the target is the card actually played by the bot in that state (imitation learning).
- Additionally, we will store the outcome of each trick (which team won the trick and how many points were gained) and the final result of the game (total points per team and winning team).

These outcome variables can later be used to define alternative labels (e.g., moves that lead to higher expected score) or for evaluation.

2.4 Limitations

A key limitation is that the dataset is based on synthetic games between AI agents rather than expert human players, which may reduce the alignment with real human strategies.

Furthermore, the chosen feature representation may not capture all subtle aspects of advanced play (for example, complex card-counting patterns or opponent modelling), leading to an incomplete description of the game state.

2.5 Potential Bias

Since the data is generated from a finite set of advanced bots, the model will tend to learn and reproduce their specific style and biases.

If the bots share similar heuristics, the dataset may contain repeated patterns and predictable behaviours, limiting diversity and causing the learned model to overfit to particular strategies.

Additionally, AI agents may lack the natural variability and occasional irrationality of human players, which can result in a policy that is strong under simulation but less robust against real human opponents.

3 Type of Machine Learning Problem

We formulate the task as a classification problem, where the model receives a representation of the current game state and must select one card to play among all legally available actions. In this setting, each decision instance corresponds to a state–action pair, and the label is the card actually played by an advanced AI bot (or another chosen policy), making the problem suitable for standard supervised learning methods.

Alternatively, the problem can be viewed as a ranking task, in which the model assigns a score to each legal card and the selected move is the one with the highest predicted score. Conceptually, this still relies on learning from examples of "good" moves but emphasises ordering actions by quality rather than predicting a single class directly.

A further extension, beyond the immediate scope of this project, would be to frame Sueca as a reinforcement learning problem, where the agent learns a policy through trial and error, optimising long-term rewards based on trick and game outcomes.

For this project, we focus on the supervised classification formulation because it is simpler to implement, fits naturally with the availability of logged self-play data, and aligns well with the course objectives. It allows us to use well-understood algorithms, clear evaluation metrics (e.g., accuracy or top-k accuracy over legal moves), and a straightforward training pipeline, while still addressing a realistic decision-making problem in a complex, partially observable game.

4 Initial Assumptions

- We assume that the game state is well captured by our chosen features (hand, current trick, trump suit, previous plays, player position and score).
- We also assume that the summary of past plays contains enough information for the model to make useful decisions, even if it does not include the full history.
- We consider that the "best move" can be approximated from the data, by using the actions of strong AI bots as a reference.
- Finally, we assume that the players in the simulations behave rationally and consistently, so that the model can learn stable patterns between game states and chosen actions.

5 Risks and Challenges

- Due to the nature of the game, since the opponents' cards are not visible, the agent would have imperfect information to work with.
- Due to the difficulty in finding real statistics of games that contain very detailed data (like trumps, game states over time, player positions), this type of data has to be generated artificially. Since data is generated by artificial means, its quality may be limited if lack of variety leads to overfitting into a few strategies, rather than more general play. This strict data may also not reflect human unpredictability and occasional irrationality.
- While not necessarily a problem, the large state space created by this problem will generate multiple possible game situations and make it more difficult for the model to generalise across those different states.
- A "correct" move will be difficult to define, since we must consider not only the move that yields the best immediate points, but also one that may lead to better future game states.